TECHNICAL PROJECT REPORT & EVALUATION

Legal Document Intelligence | Academic Submission Edition

A visual account of the system built, the measured evidence, the automated tests, engineering safeguards, deployment lessons, and the remaining evaluation gaps.

87	Automated tests passing
397	GDPR legal chunks preserved
16 / 16	Answerable benchmark questions with expected chunk in hybrid top five
1.0000	Hybrid Hit@5 on the provisional GDPR benchmark

Scope note: metrics reflect the existing saved GDPR retrieval benchmark, not general legal accuracy or independent external validation.

01 | Executive summary

Legal Document Intelligence is a modular research prototype that processes public legal PDFs and supports evidence-grounded question answering. It combines structure-aware extraction and chunking, Gemini embeddings, FAISS semantic search, BM25 keyword retrieval, Reciprocal Rank Fusion, answer generation, citation integrity checks, and a Streamlit evidence interface. The project also includes case management, provenance tracking, resumable embedding checkpoints, and API retry safeguards.

Project at a glance

Engineering evidence	Validated result
Repository inventory	61 tracked files; 46 Python files at audit
Application code	8,842 lines across 32 application Python modules
Automated testing	14 test files; 87 passed, 1 non-failing PyMuPDF warning
GDPR extraction baseline	88 pages; 397 chunks; exact legacy regression preserved
Benchmark	20 provisional questions: 16 answerable, 4 out-of-scope
Documentation	README 342 lines; architecture 309; report 426
Deployment	Public Streamlit Cloud demo; API quota may block fresh ingestion

Research question

Can a hybrid legal RAG pipeline retrieve better evidence than lexical search alone while preserving traceability and supporting cautious answer generation?

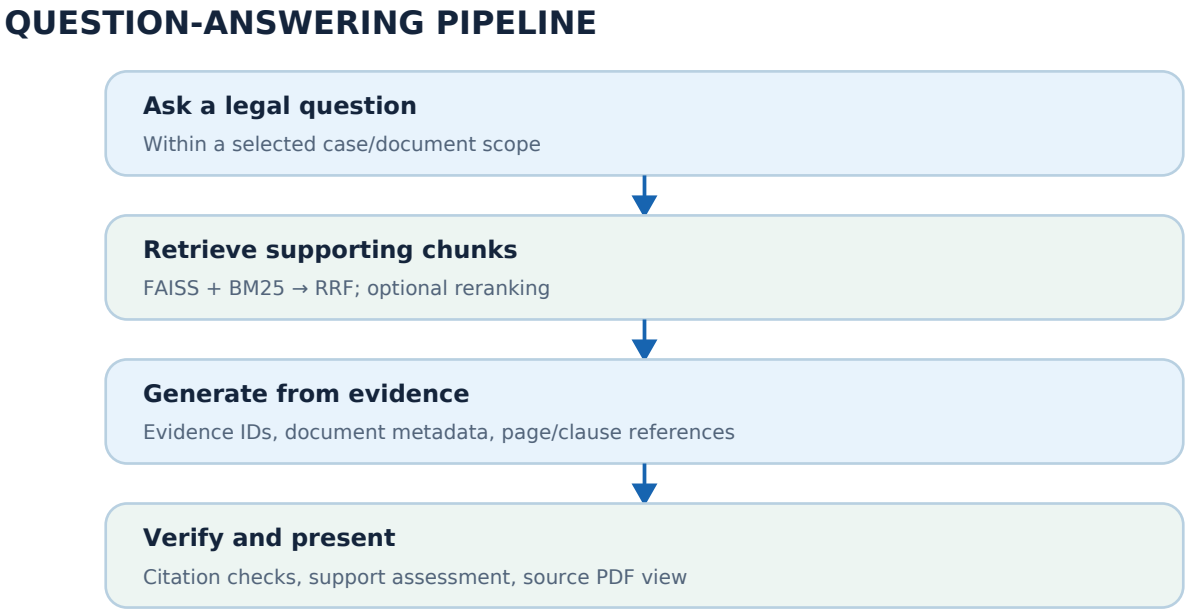
What is established

The saved retrieval benchmark demonstrates stronger top-five retrieval for hybrid than BM25 on 16 answerable GDPR questions. Automated tests verify selected reliability and integration behaviors. Comprehensive legal-answer correctness and abstention performance are not yet measured.

02 | Methodology and system components

Layer	Implementation	Purpose
Ingestion	PyMuPDF; OCR and table utilities	Read public PDFs and flag difficult pages
Legal structure	Hierarchy-aware chunking	Preserve articles, paragraphs and page metadata
Embeddings	Gemini embedding-001, 768 dimensions	Semantic vector representation
Dense index	FAISS	Semantic evidence search
Sparse index	BM25	Exact terms and clause references
Fusion	Reciprocal Rank Fusion	Merge dense and sparse rankings
Reranking	Sentence Transformers module	Optional candidate prioritization
Generation	Gemini + evidence context	Plain-language grounded response
Verification	Citation integrity + LLM verification	Check cited evidence and support
Application	Streamlit + case registry	Multi-document workspace and evidence viewer

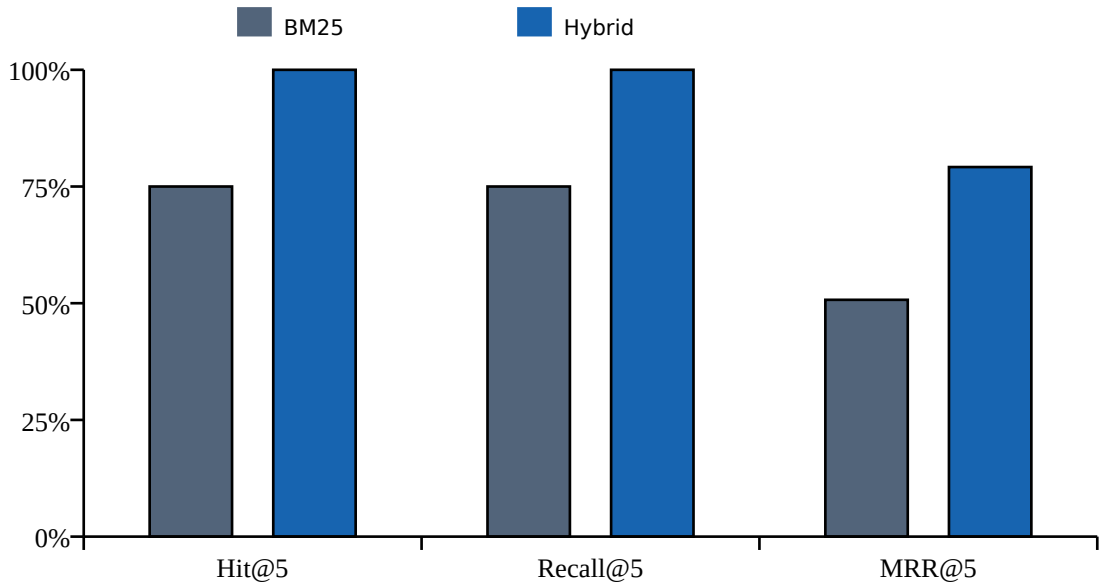
End-to-end query path



03 | Retrieval evaluation

A manually designed starter benchmark contains 20 questions based on the original GDPR PDF. Sixteen questions are answerable; four are out-of-scope and were excluded from the reported retrieval metrics. Each answerable question has one provisional expected relevant chunk.

Top-five retrieval performance



Metric (K=5)	BM25	Hybrid	Difference
Hit@5	0.7500	1.0000	+0.2500
Recall@5	0.7500	1.0000	+0.2500
MRR@5	0.5073	0.7917	+0.2844

Interpretation

BM25 found the labeled expected chunk within the top five for 12 of 16 answerable questions; hybrid found it for all 16. Four BM25 misses were recovered by hybrid retrieval. MRR@5 also increased, suggesting earlier placement of the labeled evidence. This is a small, provisional retrieval-only benchmark; it does not demonstrate answer accuracy or causal attribution to any one retrieval component.

04 | Test coverage and engineering validation

The most recent full test execution reported 87 passed, 1 warning in 6.13 seconds. The warning concerns PyMuPDF’s deprecated fitz import and did not fail the suite.

Test cases by module

Test module / area	Named test functions
Demo authentication	12
Generation and verification	12
Retrieval reliability	8
Ingestion reliability	7
Generation/verification integration	6
Streamlit progress integration	6
Embedding progress UI	6
Checkpoint recovery	5
Document identity	5
Embedding progress	3
Embedding retry limits	3
GDPR baseline	3
Pipeline progress	2
Legacy retry limit	1

The inventory lists 79 named test functions across 14 files; the executed pytest suite reports 87 passing test cases. Parametrization or other collection differences can explain this distinction; the exact collected-node breakdown was not separately audited.

05 | Reliability achievements

Baseline-preserving identity propagation

Document identity was introduced through extraction, chunking, retrieval, generation and citation verification while preserving the original GDPR baseline: 88 pages, 397 chunks, identical IDs and chunk records. A sanitization-collision test confirmed distinct document IDs remain distinguishable.

Checkpoint recovery and provenance

Embedding checkpoint files are saved atomically and validated using content fingerprints, model/dimension compatibility and document metadata. Reuse depends on the saved files surviving the runtime.

Retry delay fail-fast protection

Streamlit Cloud returned Gemini embedding retry delays of roughly 29,000–30,500 seconds. The implementation was updated to fail fast when a provider-requested delay exceeds 120 seconds, avoiding multi-hour blocking. A production screenshot confirmed the safeguard surfaced an actionable failure message.

Modular case and document management

Document and case registries, processing locks and source-aware retrieval support a multi-document legal workspace, rather than a single hard-coded PDF.

Grounded generation and citation verification

Evidence identifiers, citation metadata, integrity checks and a PDF citation viewer make source traceability a first-class system feature. Citation correctness against manually labeled ground truth remains unmeasured.

06 | Features versus evidence

Capability	Current status / evidence
PDF ingestion + hierarchy	Implemented; GDPR baseline regression verified
FAISS + BM25 hybrid	Implemented; BM25 vs hybrid K=5 metrics reported
Cross-encoder reranking	Module present; activation and performance gain not measured
Multi-document case workflow	Implemented in code; complete 2–5 PDF live demo unverified
OCR + tables	Support implemented; sample accuracy unmeasured
Bounding-box citations	Viewer functionality present; region precision unmeasured
Grounded generation + verification	Implemented; answer correctness and citation precision unmeasured
Abstention	Mechanisms present; abstention accuracy unmeasured
Live deployment	Public URL; new processing subject to Gemini quota
Test suite	87 passed at final documentation audit

Rubric alignment and evaluation gaps

The assignment requires at least 15 held-out questions, including multi-hop and at least three unanswerable cases, plus answer correctness, citation precision, abstention behavior and latency. The current 20-question benchmark is provisional; its held-out status and multi-hop composition have not been established, and the four unanswerable cases were not included in retrieval metrics. The rubric also specifies no-retrieval vs retrieval or dense-only vs hybrid as the mandatory comparison; the saved BM25 vs hybrid comparison is useful but not identical to those named options.

07 | Future work and submission priorities

Priority	Proposed improvement	Why it matters
Immediate	Reliable pre-indexed demo and cited multi-hop example	Reduces live-demo quota risk
Immediate	Held-out answer/citation/abstention/latency evaluation	Addresses required grading evidence
Near-term	Dense-only vs hybrid ablation	Matches rubric comparison wording
Near-term	Persistent cloud storage + background jobs	Survives Streamlit restarts; avoids blocking
Later	OCR/table accuracy benchmarks and region citation precision	Supports optional-feature evidence
Later	Production auth, monitoring and selective refactoring	Security and maintainability

Limitations and responsible use

This is an educational research prototype, not a legal advice system. Model output must be checked against cited source documents. API quota failures, ephemeral hosting, OCR/table extraction errors, incomplete evaluation, and demo-level authentication limit production readiness.

Conclusion

The project demonstrates an end-to-end hybrid legal RAG architecture with strong emphasis on provenance, recoverability and source-grounded outputs. Its clearest measured result is the improvement from BM25 to hybrid retrieval on a small GDPR benchmark. The extensive automated test suite and explicit operational safeguards are important engineering contributions. Future evaluation should establish answer correctness, citation precision, abstention accuracy, latency and generalization to more document types.

Project resources

Repository: https://github.com/vinnyyxd/legal-document-intelligence
Demo: https://legal-document-intelligence-demo.streamlit.app/
Detailed methodology: data/evaluation/README.md
System design: docs/ARCHITECTURE.md
Technical write-up: docs/PROJECT_REPORT.md